

Regional-scale bed roughness beneath ice masses: measurement and analysis

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Abstract

We outline a method of assessing the roughness of the bed beneath ice-masses, applicable to data collected in a wide variety of (glaciated) environments. Roughness at scales $> 10^2$ m in particular is poorly constrained and understood in glaciated systems, and regional geophysical surveys collected over the past 30 years are ideally suited to addressing this issue. We demonstrate the technique using a radio-echo sounding data set collected during the 1970s, showing that regional characterization of roughness over a wide range of wavelengths (1 to > 40 km) is possible using Fast Fourier Transforms, even with quite coarse horizontal sampling. Coarse sampling captures all of the variation of higher resolution analyses and the technique appears robust up to the limit imposed by the Nyquist frequency.

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1. Introduction

Characterising the bed beneath ice masses, both present and former, is an important first-order measurement in understanding the potential effects of the bed on ice dynamics at horizontal scales $> 10^2$ m. Whilst this has been addressed qualitatively in the case of ice streams (e.g. the Siple Coast ice streams; Shabtaie and Bentley, 1987) and, at least partially, theoretically (Hindmarsh, 1998), a quantitative regional-scale analysis has yet to be done. Indeed, roughness is often ignored completely (cf. Bennett, 2003, p. 321). Initially, the lack of analysis was due to methodological difficulties and the lack of data at an appropriate resolution (cf. Rose, 1978). This is no longer the case and a quantitative analysis of bed roughness at meaningful scales is possible with extensive

ice-penetrating radio-echo sounding (RES) data, where the bed is imaged, and sonar and echo-sounding data from the sea floor. This paper outlines the rationale and a methodology for deriving regional-scale bed roughness, with reference to the extensive Scott Polar Research Institute (SPRI) RES data sets covering Antarctica (Rose, 1978, 1979), although the method may be applied to any geophysical survey with topographic data.

2. Background

Roughness is the degree of irregularity in a surface. It is the variation of elevation with distance and as such requires elevation relative to a datum (rather than absolute elevation) for analysis (Herzfeld et al., 2000). One result of defining roughness in this way is that it is effectively scaleless in the vertical dimension, making

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comparison between different data sets possible across similar horizontal spatial scales. Roughness has been examined quantitatively only rarely in glaciology: at the ice surface, usually with regard to satellite or meteorological analyses (e.g. Herzfeld et al., 2000); and the ice bed (Hubbard and Hubbard, 1998; Hubbard et al., 2000). Such analyses are usually performed at scales $<10^2$ m, however.

We address roughness at the scale 10^2 – 10^5 m. This spatial scale is significant because it includes a range of glacial bedforms that appear at a scale of 10^2 – 10^4 m (including drumlins, outwash channels, and linear bedforms) (e.g. Ó Cofaigh et al., 2002; Stokes and Clark, 2003), and thus may shed light on the processes, which form them. Fluvial channels, larger hills, valleys, and canyons can also appear at scales greater than these. As well as the importance of bed forms, roughness at the scale $>10^2$ m varies as the expression of bedrock, reflecting differential erosion, faulting, and changing lithology.

Although the links between bed roughness and ice dynamics have been treated more extensively at the $<10^2$ m scale, bed roughness is often invoked as a loose mechanism for explaining regional elements of ice dynamics, but with only weak mechanistic understanding. For example, relative changes in roughness are ascribed a role in changing basal temperature and ice dynamics in northeastern Sweden during the last glacial cycle (Hättestrand and Stroeven, 2002). Such a role is invoked through topography-induced strain heating of the ice mass, especially at scales close to the icethickness. Pattyn (1999) uses a similar mechanism to change ice dynamics in areas of more mountainous topography in Dronning Maud Land, where basal temperature is affected strongly by sharply varying topography. Furthermore, ice-streaming areas are qualitatively described as having very low roughness, with elongate bedforms and an overall 'smoothed' appearance, especially in the downstream direction (Stokes and Clark, 2001, 2003). Therefore, roughness and ice dynamics are inherently linked (or coupled) at the $>10^2$ m scale.

Roughness is a function of landform and substrate, together with ice dynamics, in a feedback loop; fast-flowing ice deforms or erodes the bed and is typically at the pressure-melting point, allowing the existence of water at the bed and substrate conditions conducive to fast-flow. Observations of fast-flow associated bedforms over a variety of hard and soft lithologies suggest, however, that substrate type may turn out to be less significant than previously thought in controlling fast-flowing ice (Ó Cofaigh et al., 2002; Stokes and Clark, 2003).

The overall effects of regional-scale roughness on ice dynamics may be broadly described as tractional, with associations to frictional heating, resistance, and hydrological organisation. These factors are brought together

by the 'sticky-spot' hypothesis, for example (Alley, 1993). Hättestrand and Stroeven (2002) suggest that strain heating may provide a direct link to ice dynamics. Hindmarsh (1998) finds that changes in bed elevation and ice thickness (approximately the same thing where bed elevation changes more rapidly than surface elevation, as is often the case) can have important impacts on effective pressure at the ice bed through hydrological influences. These processes are understood only poorly, however, and establishing the character of the bed beneath ice masses regionally and at scales of 0.5 – $10 \times$ icethickness (Hindmarsh, 1998) is critical to furthering the debate.

3. Data availability and quality

In this paper we use Antarctic RES from the region around Siple Coast to demonstrate the roughness quantification technique presented (Fig. 1). However, the methodology is intended to be applicable to any large-scale geophysical trackline survey data (radar, sonar, or radio-echo sounding), airborne or ground-based, which include bed returns plotted with respect to an absolute datum. These data may have been collected digitally or be digitized from paper or film archives. Although errors from digitization are obviously avoided using modern digital recording techniques, this is not possible with many, older data sets and to reject data on this principal would make much potentially useful, archived data redundant. Many presently and formerly glaciated environments, terrestrial and marine, have been surveyed and a wide range of data is potentially available.

Good quality navigation is desirable with any data set. However, the absolute errors in the position of any one data point are less relevant when considering properties along survey lines than the accuracy of the relative placing of preceding and following data points. Errors in the latter are typically orders of magnitude less than those associated with absolute positioning and, thus, will not significantly affect any calculation of along-track properties. This is particularly significant when considering data collected before the advent of the global positioning system (GPS).

Migration of geophysical survey data is common to correct reflector positions, improve horizontal resolution, by focusing energy, and collapse diffractions in the record from high relief or discontinuities (Kearey and Brooks, 1991). Most of the geophysical data archive for glaciated regions is unmigrated. However, whilst this may hide some fine detail in the bed reflector, lack of migration is not necessarily a terminal problem. Firstly, roughness calculations using unmigrated data are minimum estimates of this property. Secondly, short wavelength roughness is not added anywhere in the

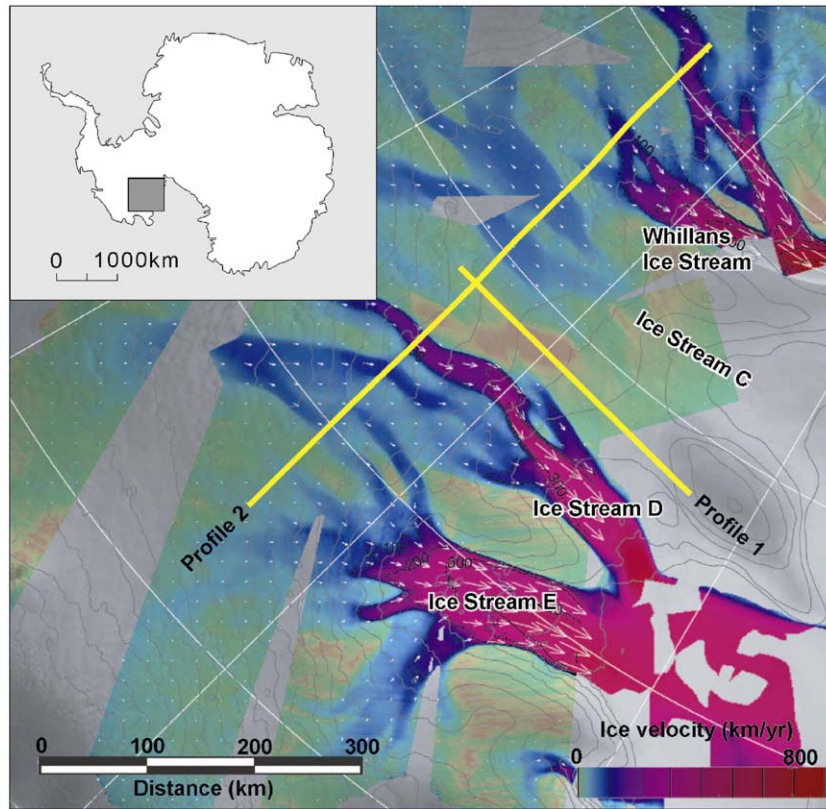


Fig. 1. Location of SPRI radio-echo sounding transects over Siple Coast region of West Antarctica. Location of profiles referred to in Figs. 5 and 6 are noted. Satellite-derived ice-surface velocities outlining position of ice streams are taken from Joughin and Tulaczyk (2002). A key showing colour-coded ice surface velocities is also provided.

processing. Finally, the lack of migration is a form of smoothing, removing sharp transitions which make the bed records easier to model statistically (although it should be acknowledged that some information may be lost).

The data used here by way of example are available as unmigrated, time-continuous recordings of 'A-scope' data assembled on film as 'Z-scope' output. The 'Z-scope' records are marked at 20 s (time) intervals (approximately every 2 km). These data, extending more than 400,000 km, cover most of West Antarctica outside of the Antarctic Peninsula and about a third of East Antarctica, east of Dome A and Wilkes Land (Fig. 1). As well as the original film data, a crude digital version of the bed data set is available, used in the compilation of the BEDMAP database (Lythe et al., 2000). These data were originally digitized at the 20 s time markers using a semi-automatic trace reader, corrected, and subsequently tied with navigation records (Rose, 1978).

Navigation control on these data was through an inertial navigation system adjusted with numerous navigation fixes. Errors in absolute position are reported as normally <5 km, with individual errors rarely

exceeding 10 km (Rose, 1978). In a flightline of 200 km, this represents an error of <10% in absolute position at the end of the line, although the largest errors typically occur towards the middle of any flightline. Absolute errors in the relative positioning of datapoints are estimated as <50 m.

The overall quality of the data is excellent; 86% of the bed is imaged. The missing bed returns are typically caused by extremely thick ice or obscuring by surface crevassing. The vertical resolution of the data is also good. We estimate vertical errors are in the order of ± 25 m, resulting from a combination of digitizing errors, velocity corrections, and corrections to aircraft altitude measurements (Rose, 1978; Drewry, 1983).

4. Data preparation

Straight sections of continuous data are required for further analysis. This is rarely the case in many geophysical data sets and, therefore, the data must be cleaned, and then checked for orientation and regular sampling interval. Cleaning the data set is done using

linear interpolation to fill small data gaps (for the purpose of establishing large-scale roughness characteristics). Siegert et al. (in press) use linear interpolation to fill data gaps no larger than 7 km. Such pre-processing avoids adding short-wavelength roughness at the scales of interest. Subglacial bed elevation is often calculated as a function of surface elevation and ice thickness. Changes in sampling direction due to turns in survey lines may produce artificial variations in ice thickness, for example. Directional changes in survey lines require that the lines therefore be broken. Where the survey forms a grid or features lines of similar orientation, only lines of the same direction are directly comparable.

Fourier-based analyses of roughness require a fixed sampling interval. Most geophysical data, however, are sampled regularly with respect to time, not distance. Thus, the regularity of datapoint spacing also needs to be checked. First, the mean or nominal sampling interval is determined, and the actual gap between each point is calculated. Where the actual gap is less than the nominal, datapoints may be deleted (if necessary) to create the appropriate systematic gap. Similarly, datapoints may be inserted through linear interpolation where the gap significantly exceeds the nominal distance. Ultimately, the degree to which this is done is somewhat subjective, but if the mean and nominal sampling intervals are the same and the standard deviation is small, then the assumption that the actual and nominal sampling intervals are the same for all datapoints is a fair one.

The crude digitized Antarctic RES data set is the initial basis for this example analysis. A maximum of 4 consecutive data points were interpolated where required; a notional distance of approximately 8 km. The amount of interpolation is minimal, however. Less than 5% of the 5328 data points were involved in interpolation, and 82% of this was performed over only 1 data point, and 99% over 3 or less data points. The orientation of the Siple Coast data set was originally planned to lie orthogonal and parallel to the direction of ice flow (Rose, 1978). This allows the grid to be broken down into compound flightlines of flow-parallel and flow-perpendicular data. Data outside of these directions were rejected.

The actual mean interval between data points is 2.21 ± 0.24 km. There was no requirement to add extra data points to the data set, but where the gap between consecutive data points was significantly lower than 2.21 km, points were removed until the correct sampling interval was created. The data set thus has a notional sampling interval of 2.21 km. The errors in using the assumed rather than actual sampling interval equate to an average $6.3 \pm 4.6\%$ at the end of a line over an average distance of 180 km ($n = 66$). The size of the error is not significantly correlated with the overall length of any line; that is to say longer lines do not have larger

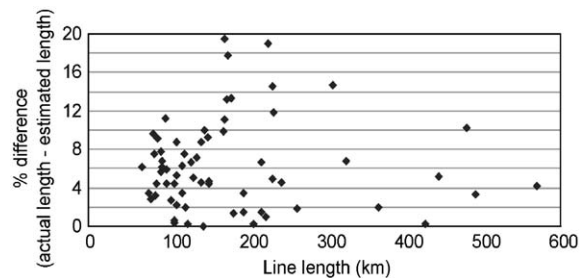


Fig. 2. Graph showing percentage error in length of RES transect (using assumed sample interval of 2.21 km) versus actual line length.

errors (Fig. 2). The small size and lack of systematic variation in the error thus derived indicate that the sampling interval assumption is a valid one. This sampling interval gives a Nyquist frequency equivalent to ~ 4.5 km. With an average ice thickness in the Siple region of 1.38 ± 0.57 km, derived from the SPRI RES data set, the Nyquist frequency is approximately 3 ice thicknesses. This is a possible limitation of the data, but one that is addressed below.

5. Roughness calculations

We apply a Fast Fourier Transform (FFT) in analysing the RES data set further. In order to meet the assumptions for this analysis, the data must be normally distributed about the mean and show second-order stationarity; that is the mean is constant and independent of location, and covariance is dependent on the distance between points but not their absolute location. Lack of migration in geophysical data works to an advantage here because the lack of extremely sharp gradients reduces the impact of both Gibb phenomena and the artificial insertion of high frequency components in fitting to discontinuities in the data (cf. Solymar, 1988). Where sharp bed transitions are encountered, it may be necessary to break the line.

The most readily used form of FFT analysis requires lines with 2^N datapoints. Exact sampling line length is determined by this constraint. A suggested minimum value of $N = 5$ give lines 32 datapoints long. This represents the minimum length required for our investigation. We do not choose longer lines, as we would wish to retain the local nature of the investigation. Other users of the technique may, however, wish to expand the length of the sections to be processed, but should we aware that the regional nature of the investigation may suffer as a consequence.

Lines of less than 32 datapoints are removed in pre-processing. Each 2^N datapoints are used to characterize roughness for the central point. There is thus a

roughness data point for each moving 2^N segment along a survey line. Moving 2^N datapoint blocks are extracted and detrended using a least-squares linear regression, to remove the very long-wavelength components (e.g., Fig. 3A). Units are converted to be the same in both axes, e.g. metres for both distance and detrended elevation. The data are finally passed through a forward Discrete Fourier Transform (FDFT) using a rectangular window. For a data set $x[n]$ with index n in the range $0 \leq n \leq NT-1$, where NT is the total number of data points, the FDFT is defined to be:

$$X[k] = \sum_{n=0}^{NT-1} x[n] \exp(-i2\pi F_k n), \quad (1)$$

where

$$F_k = \frac{k}{NT}. \quad (2)$$

We undertake amplitude normalization, to reveal the true amplitudes in the original data set, by dividing the amplitudes of the AC component of the FFT by $NT/2$. A single roughness value is calculated as the integral of the spectral power density plot. Integration is carried out automatically using the trapezoidal rule between all adjacent power density values. Roughness is calculated over specific wavelengths, characterised in a

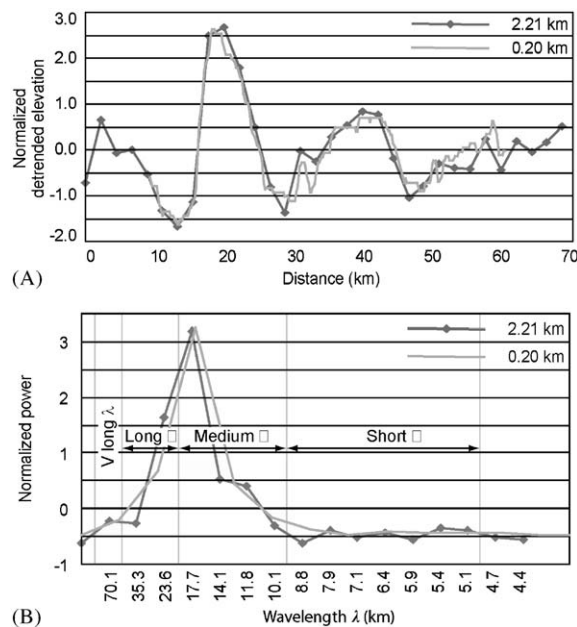


Fig. 3. Examples of power spectra. (A) Profiles at 2.21 km and 200m sampling interval used to generate respective power spectra. (B) Power spectra calculated at one point in Profile 1. Note how normalised powers for sample intervals of both 2.21 and 0.2km are similar. Note also how bulk of total power occurs under wavelengths greater than ‘medium’.

Table 1

Definition of roughness wavelengths referred to in this paper

Wavelength class	Roughness wavelength (fraction of line length)
Short	0.0625–0.125 (4–9 km)
Medium	0.125–0.25 (9–18 km)
Long	0.25–0.5 (18–35 km)
Very long	>0.5 (>35 km)

continuum from very long to short wavelengths (Table 1). However, the choice of wavelengths is partially determined by the frequency output of the FFT. Wavelength is the reciprocal of frequency and describes an approximate power curve (Fig. 3). Therefore, in order to integrate over a similar number of intervals and get useful variation in power, the range of wavelengths in any one category is somewhat limited. In our case, wavelength classes are divided at approximately 0.5, 0.25, 0.125, and 0.0625 sample line length (giving in our example, wavelength classes of ~35, 18, 9 and 4.5 km, respectively). For other situations (for other lengths of data) alternative wavelength class strategies may need to be developed (which is simple to arrange).

The SPRI RES data are normally distributed about the mean and show second-order stationarity. We follow Rose (1978) and use ~50 km segments of data to calculate roughness, the roughness of a single point being defined by the character of the bed for 25 km in either direction. 2^5 datapoints gives an actual sample length of $32 \times 2.21 = 71$ km. Data blocks retain their second order stationarity after being detrended (Fig. 3). The output frequencies available, and their division into the wavelength classes, are shown in Table 2 and Fig. 3 for a ~50 km line with both a 2.21 and 0.2 km sampling interval.

Roughness power at short and medium wavelengths is relatively insensitive to both the position and size of the integrating window (Fig. 4). The sensitivity of wavelength power to definition becomes greater towards longer wavelengths (Fig. 4C, D, G, and H), but this is a result of both the greater overall levels of power, especially at 17.7 km in this case (Fig. 3B), and the relatively small number of wavelength observations used to calculate the parameter. The increased sensitivity with longer wavelengths is not that important, however, because the patterns of variation are largely the same (Fig. 4) and it is really only the shorter wavelengths (<20 km) that are of glaciological significance (see above).

The overall result is a series of values of each roughness component for the midpoints of each moving 50 km data-block. The analysis captures the scales of variation along the bed extremely well in both

Table 2
Frequency and wavelength classifications

Frequency	Wavelength ($0.001/f$, km)	Wavelength class	
0.0000000			
0.0000141	70.720	> 40	V long
0.0000283	35.360	20–40	Long
0.0000424	23.573	20–40	Long
0.0000566	17.680	10–20	Medium
0.0000707	14.144	10–20	Medium
0.0000848	11.787	10–20	Medium
0.0000990	10.103	10–20	Medium
0.0001131	8.840	5–10	Short
0.0001273	7.858	5–10	Short
0.0001414	7.072	5–10	Short
0.0001555	6.429	5–10	Short
0.0001697	5.893	5–10	Short
0.0001838	5.440	5–10	Short
0.0001980	5.051	5–10	Short
0.0002121	4.715	< 5	Not used
0.0002262	4.420	< 5	Not used

flow-parallel (Fig. 5) and flow-perpendicular (Fig. 6) lines. For example, the growth and dominance of roughness at the scale of 10–20 km at between 120 and 180 km along-track in Profile 1 is imaged well (Fig. 5). This profile is located immediately adjacent to Ice Steam D and shows well the downstream removal of all wavelength roughness elements. Profile 2 is approximately flow-perpendicular and again shows systematic variation of roughness along-track (Fig. 6). Areas of ice streaming are generally smoother than those outside, for example. If completed for a whole survey, a full spatial picture of roughness in flow-parallel and flow-perpendicular directions may thus be built up, e.g. Siegert et al. (in press).

Roughness does not vary equally at all scales along a line. This is visible in correlation matrices of the various elements of the power spectra (Table 3). Roughness at any particular scale is most strongly correlated with the next nearest scales up and down. Correlation drops off strongly with increased scale difference in roughness, for example the very low and

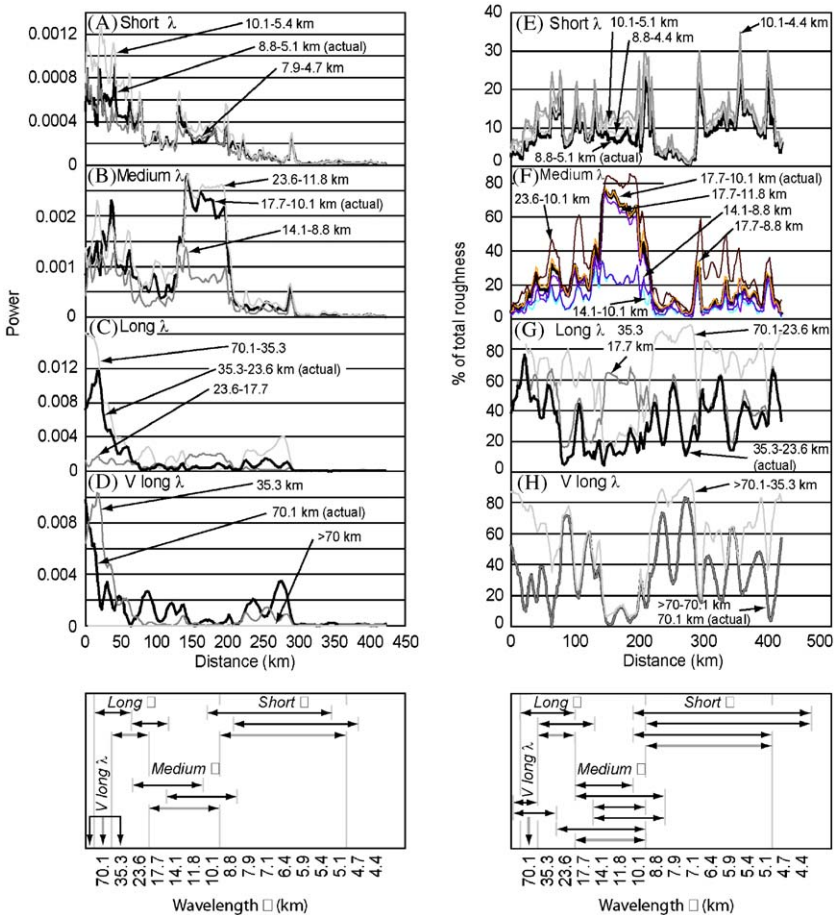


Fig. 4. Variation in power along Profile 1 with respect to definition of wavelength window. Windows are varied according to position (A–E) and size or number of wavelength observations within a window (F–J).

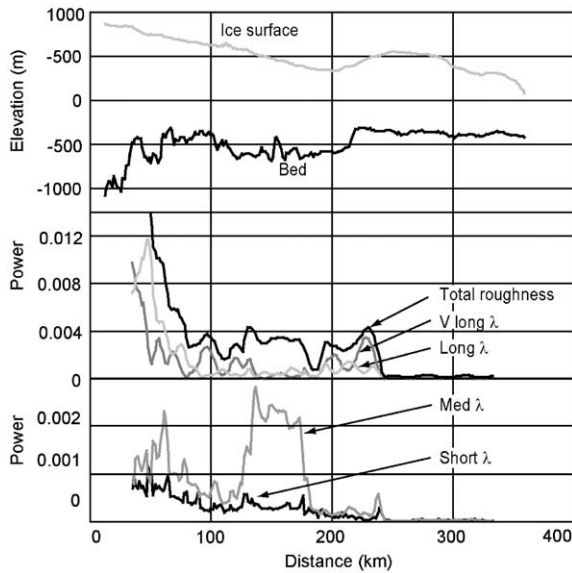


Fig. 5. Roughness data from a flow-parallel transect (Profile 1; Fig. 1). Roughness classes are as detailed in Table 1. Note that all roughness components decrease in power downflow, but their relative importance varies strongly.

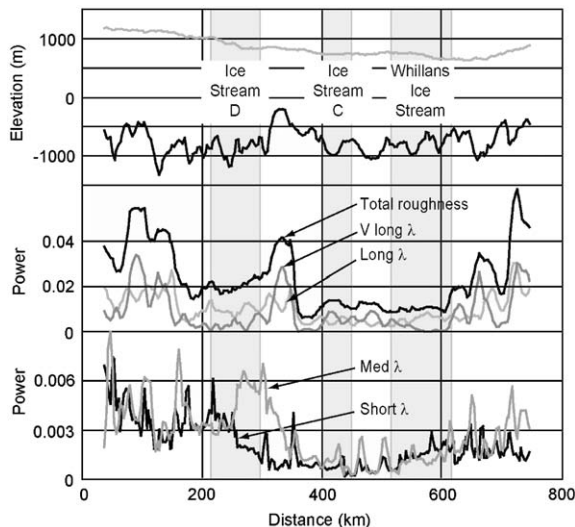


Fig. 6. Roughness data from flow-perpendicular profile 2 (Fig. 1). It should be noted that this profile crosses several of major ice-streams in Siple Coast region.

insignificant correlation between short and very long wavelengths in Profile 2. Medium-wavelength roughness is most strongly correlated with short-wavelength roughness, whilst generally long and very long wavelength roughnesses are also very highly correlated.

6. Sampling resolution and sub-ice thickness variations

Many of the processes, which potentially relate bed roughness to ice flow are thought to operate at wavelengths of a few ice-thicknesses or less (e.g. Lliboutry, 1993). The Nyquist frequency for a ~ 2.21 km sampled data set such as this is 4.5 km, or close to the boundary where bed roughness is traditionally thought to have an impact on ice dynamics, as noted above. Does this sampling resolution therefore adequately capture variation at higher frequencies?

In order to address this question, we returned to two flow-parallel and two-flow perpendicular examples of the original RES records. The bed on these records was re-digitized at a much higher resolution and tied to the existing 2.21 km data set through the 20 s marks, which also provide geographic coordinates. Along-track resolution of the scans is calculated in the order of 30 m per pixel (geographic distance between tie points divided by number of scan pixels), whilst vertical resolution is approximately 5 m, well within the errors of the data set. The records were subsampled to a constant 200 m interval (average actual sampling interval 204 ± 25 m) and geographic coordinates extrapolated from individual tie points.

The output of the FFT analysis with this level of data allows assessment of the power spectrum down to wavelengths of 1 km. The high-resolution data show the same patterns of roughness along flightlines as the 2.21 km interval data set (Figs. 3 and 7). The proportions of power at any particular wavelength λ seen in the higher sampled data set are imaged as clearly in the 2.21 km interval data, giving strong confidence in the capability of the lower resolution data to represent roughness along a flightline close to the Nyquist frequency. Furthermore, the correlation matrix of wavelength components for the 200 m interval data show that wavelengths < 10 km are strongly correlated, even down to 1 km (Table 4). We can thus demonstrate that the 2.21 km data set short wavelength component is representative of roughness to a level below a single ice-thickness.

7. Discussion

Are these roughness calculations meaningful in any glaciological context? We have already noted that roughness at the scale $> 10^2$ m may be linked to ice-mass behaviour through strain heating at the base, or a combination of frictional-heating, resistance, and hydrological organization (the “sticky-spot” hypothesis). We have also seen that there is a possible division into short/medium and long/very long-wavelength roughness patterns of behaviour. It is particularly significant to note that short ($< 10^4$ m) and longer-wavelength

Table 3

Roughness power correlations for both profiles, located in Fig. 1

Pearson 'r' values for inter-wavelength correlations

	Profile one (flow-parallel), $NT=185$			Profile two (flow-perpendicular), $NT=239$		
	Med.	Long	V long	Med.	Long	V long
Short	0.60	0.76	0.54	0.53	0.35	0.03
Med.		0.28	0.12		0.47	0.19
Long			0.71			0.54

	Flow-parallel (all data), $NT=3247$			Flow-perpendicular (all data), $NT=3269$		
	Med.	Long	V long	Med.	Long	V long
Short	0.85	0.70	0.40	0.74	0.73	0.60
Med.		0.77	0.47		0.70	0.68
Long			0.68			0.72

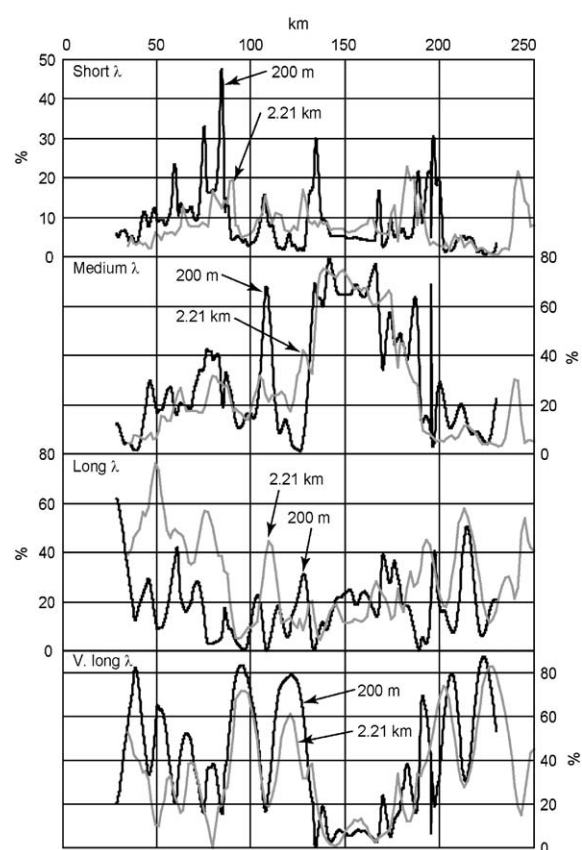


Fig. 7. Wavelength components for profile 1 at long (2.21 km) and short (200 m) horizontal sampling intervals. Powers are expressed as a percentage of total power of that data point.

roughness ($>2 \times 10^4$ m) is weakly related. This is a sensible distinction in the landscape, perhaps marking the transition between 'landforms' and bedrock-

related variation. However, in any glaciated environment, there is also the question of pre-existing topography, unrelated to glaciological processes. We are unable to address pre-existing topography though, except to note that glaciological processes at the large-scale often tend to emphasise or preserve geomorphological structures which pre-date glaciation (e.g. Delisle, 1997).

The close correlation of wavelength components <10 km in the 200 m interval data set is intriguing because it suggests that patterns of roughness below this scale are more or less equal. This may also be because this is the scale at which landforms are expressed at the surface, and there is an inherent link between landforms and glaciological setting in areas influenced by large ice-masses.

Siebert et al. (in press) analysed the bed roughness of the Siple Coast ice streams, using the technique presented in this paper applied to the full SPRI RES data set. The analysis revealed three distinct features, which relate bed roughness to ice dynamics in this region. First, bed roughness was found to decrease downstream, as illustrated in Fig. 5. If the bed morphology cannot be modified upstream, this may limit the inland migration of the ice streams. Second, with the exception of Ice Stream C, the beds of inter-stream regions were found to be rougher than beneath the ice streams (Fig. 6). This may inhibit the lateral migration of these ice streams. Third, in contrast to other ice streams, the bed beneath the inter-stream regions bordering Ice Stream C, which is currently inactive, was found to be smooth at all scales. Consequently, when the ice stream was active, there would have been little opposition from the bed to the lateral spreading of this ice stream.

Quantification of subglacial roughness, using techniques presented here, may have a powerful role to play in

Table 4

Correlation matrix of wavelength components for RES data sampled at a 200 m interval

Wavelength component (km)	Pearson 'r' correlation matrix					
	2–5	5–7	7–10	10–20	20–40	>40
1–2	0.568	0.405	0.461	0.281	0.217	–0.008
2–5		0.458	0.527	0.328	0.274	–0.021
5–7			0.385	0.390	0.306	–0.108
7–10				0.378	0.360	0.142

understanding ice sheet processes. One interesting potential avenue of investigation involves quantifying formerly glaciated terrain (both terrestrial and marine) from existing geophysical and satellite techniques. Such quantification would allow modern subglacial terrain to be compared with former landscapes, which may help in the reconstruction of former ice sheet dynamics and landscape evolution processes.

8. Conclusions

We have described a methodology for defining the roughness of the bed of ice masses from geophysical data, illustrated using an Antarctic RES data set. The method gives spatially robust results over a range of wavelengths, even at relatively coarse sampling intervals. Roughness at particular wavelengths may be related to a variety of glaciological phenomena, where the theoretical and observational have yet to converge fully. Examining roughness at a regional scale using these methodologies is therefore important in characterizing further how ice may interact with its bed, in particular at scales of several ice-thicknesses. Furthermore, the method has wide application to sonar and echo-sounding data collected from flooded glaciated shelf environments, which have been surveyed extensively in the recent past, as well as forming a potentially powerful tool for the quantitative comparison of glaciated regions.

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